



DIGITAL IMAGE PROCESSING

Lecture 5

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Chapter 3: Intensity Transformation and Spatial Filtering



Spatial Operations

- Single-pixel operation (Intensity Transformation)
 - Negative Image, contrast stretching etc.
- Neighborhood operations
 - Averaging filter, median filtering etc.
 - Geometric spatial transformations
 - Scaling, Rotation, Translations etc

Image Enhancement

- The objective of image enhancement is to process an image so that the result is more *suitable* than the original image for a specific application.
- There are two main approaches:
 - Image enhancement in spatial domain: Direct manipulation of pixels in an image
 - Point processing: Change pixel intensities
 - Spatial filtering
 - Image enhancement in frequency domain: Modifying the Fourier transform of an image

Some Basic Intensity Transformation Functions

- Image Negatives

$$s = L - 1 - r$$

- S is the output intensity value
- L is the highest intensity levels
- r is the input intensity value
- Particularly suited for enhancing white or gray detail embedded in dark regions of an image, especially when the black areas are dominant in size

Some Basic Intensity Transformation Functions

- Image Negatives



Some Basic Intensity Transformation Functions

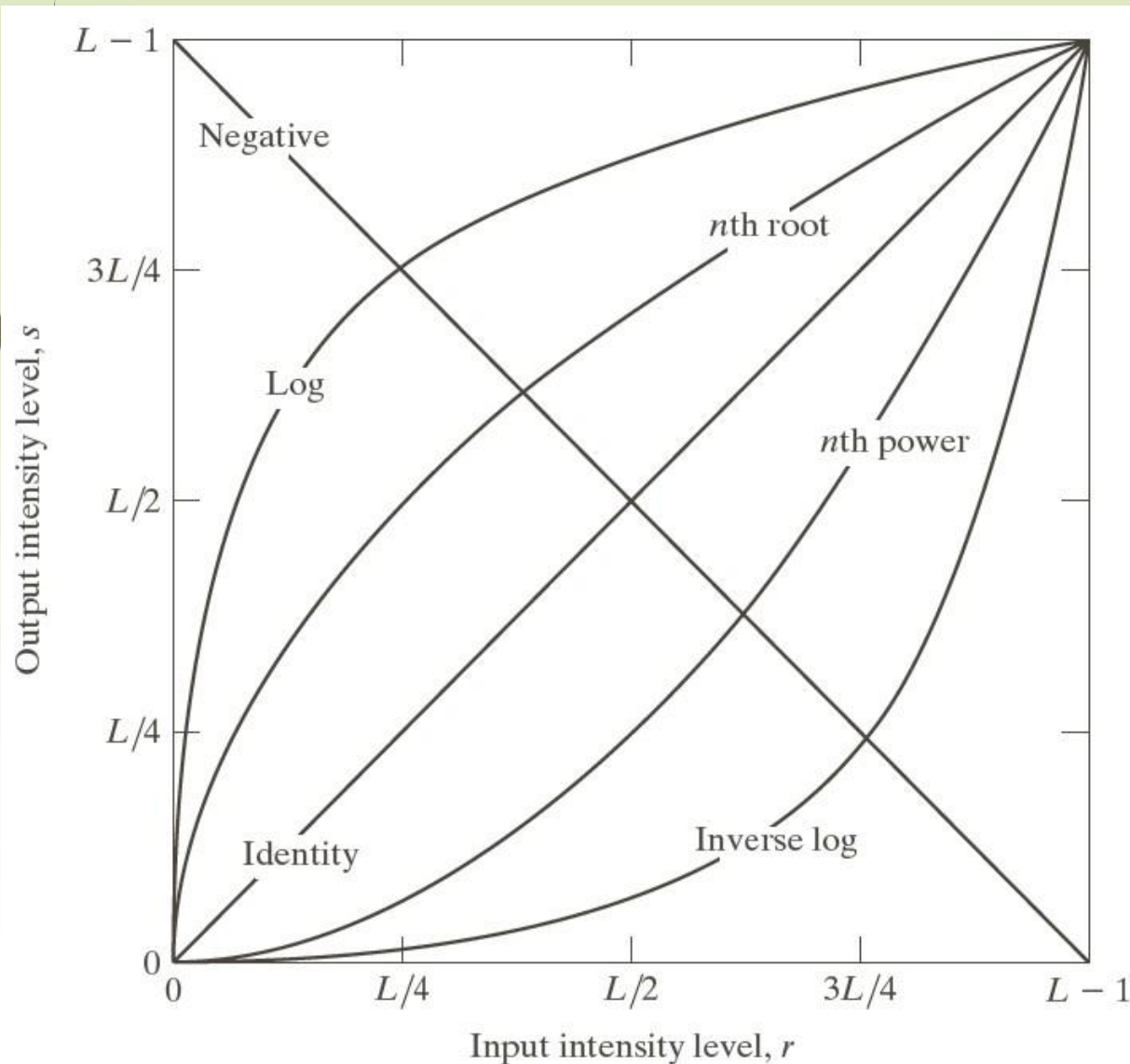


FIGURE 3.3 Some basic intensity transformation functions. All curves were scaled to fit in the range shown.

Some Basic Intensity Transformation Functions

- Log Transformations

- $s = c \log(1 + r)$

- c is constant

- It maps a narrow range of low intensity values in the input into a wide range of output levels

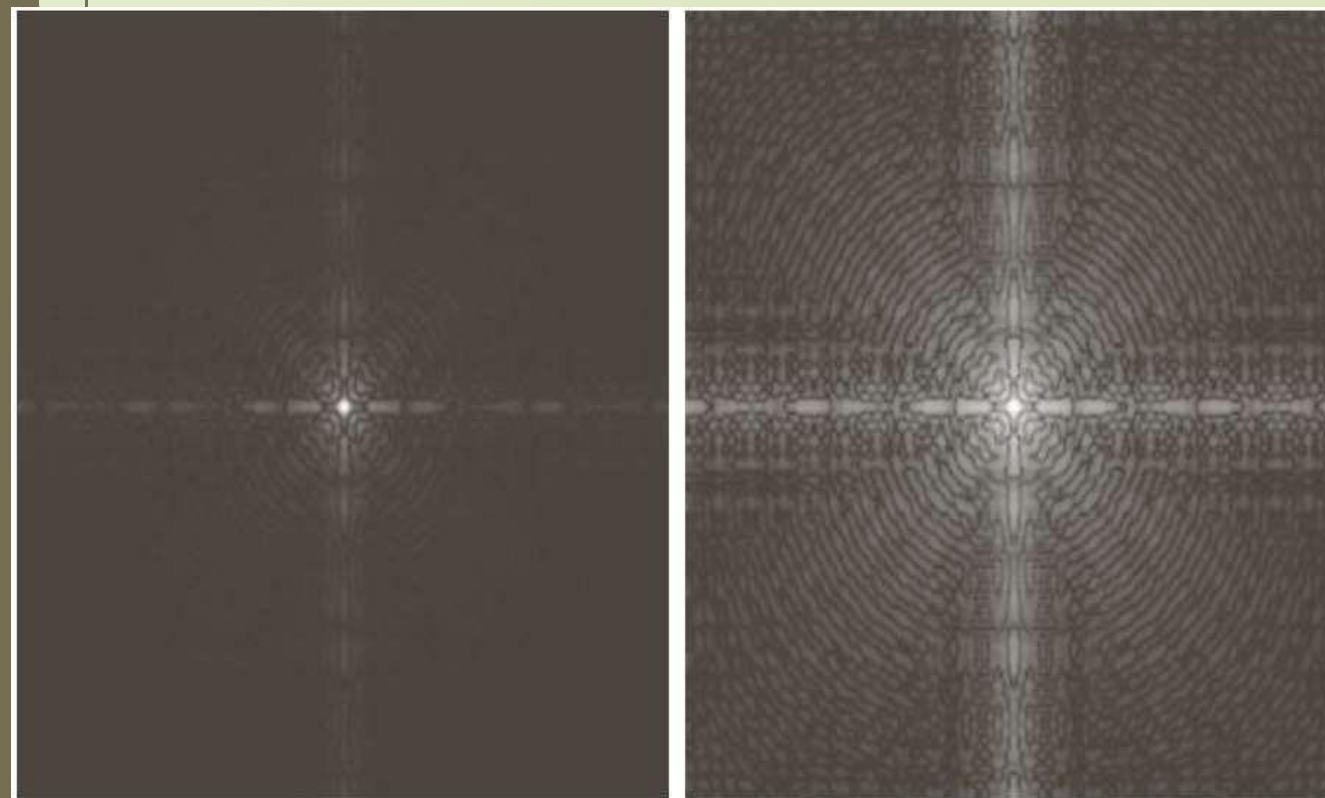
- The opposite is true of higher values of input levels

- It expands the values of dark pixels in an image while compressing the higher level values

- It compresses the dynamic range of images with large variations in pixel values

Some Basic Intensity Transformation Functions

- Log Transformations



a b

FIGURE 3.5

(a) Fourier spectrum.

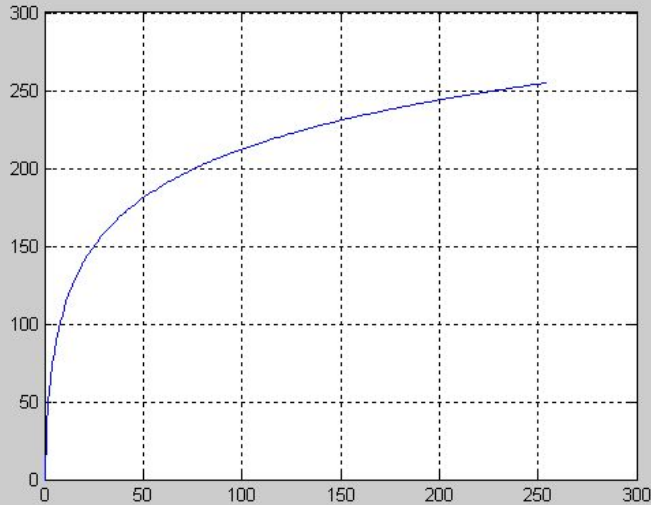
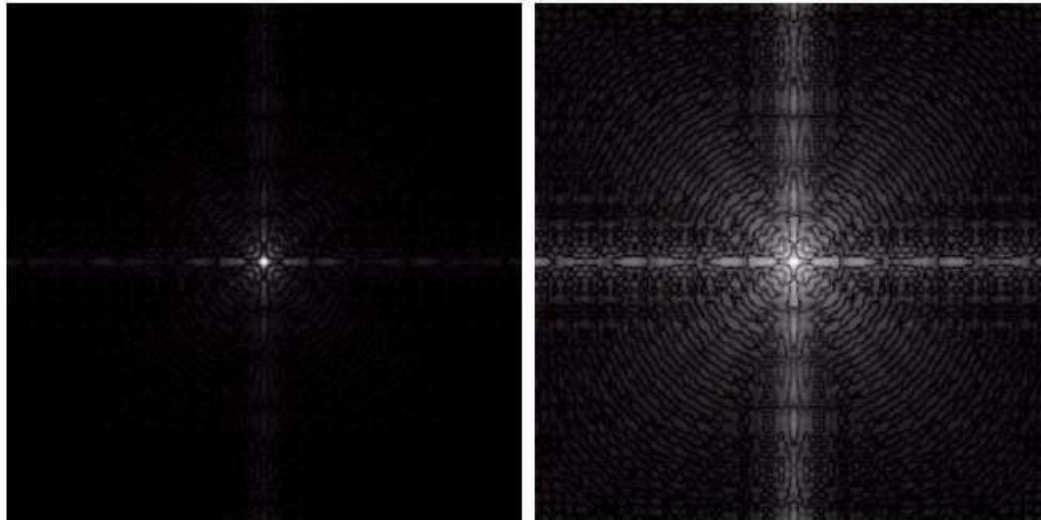
(b) Result of applying the log transformation in Eq. (3.2-2) with $c = 1$.

Log Transform

a b

FIGURE 3.5

(a) Fourier spectrum.
(b) Result of applying the log transformation given in Eq. (3.2-2) with $c = 1$.



$$T(r) = c \log(1 + r)$$



Some Basic Intensity Transformation Functions

- Power Law (Gamma) Transformations

- $s = c r^{\gamma}$

- c and γ are both positive constants

- With fractional values ($0 < \gamma < 1$) of gamma map a narrow range of dark input values into a wider range of output values, with the opposite being true for higher values ($\gamma > 1$) of input levels.

- $C = \gamma = 1$ means it is an identity transformations.

- Variety of devices used for image capture , printing, and display respond according to a power law.

- Process used to correct these power law response phenomena is called *gamma correction*.

Some Basic Intensity Transformation Functions

- Power Law (Gamma)

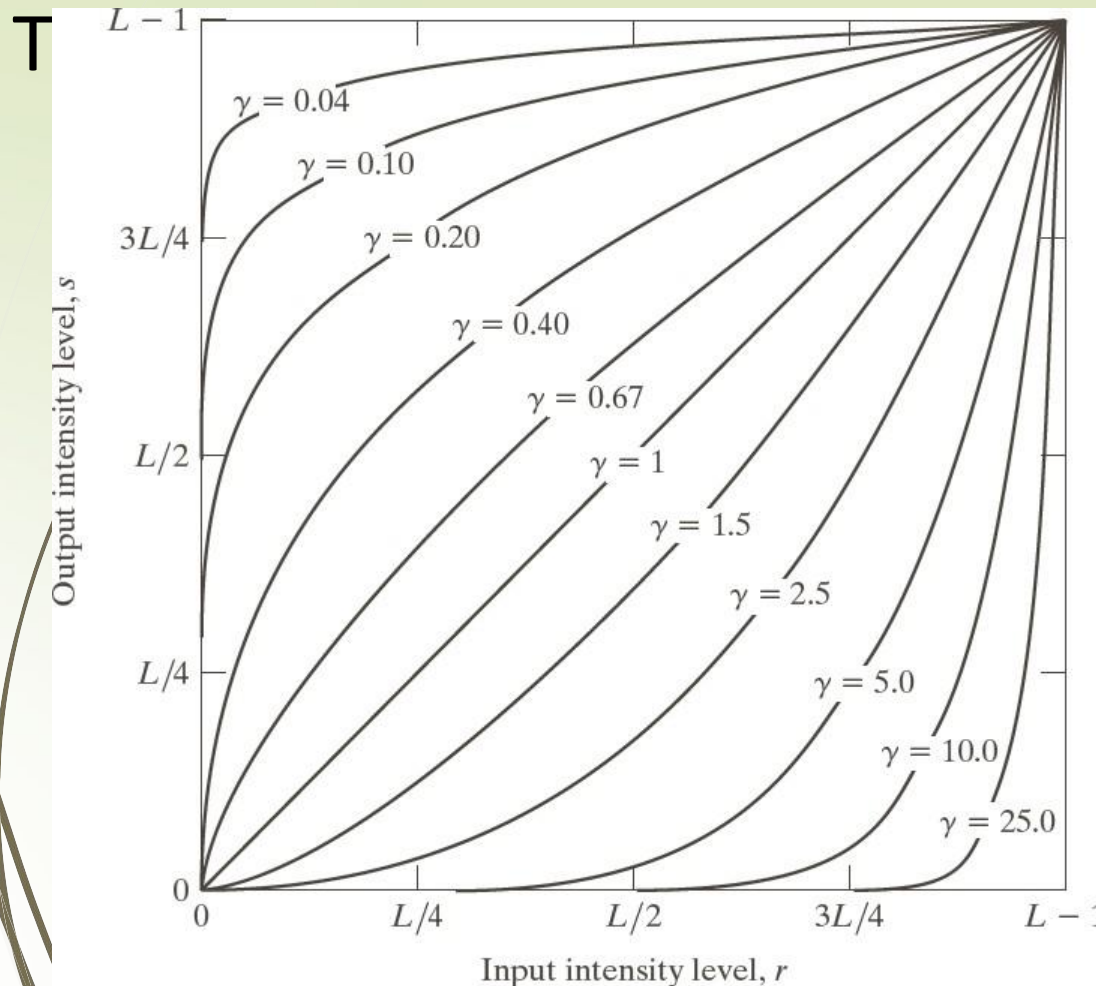
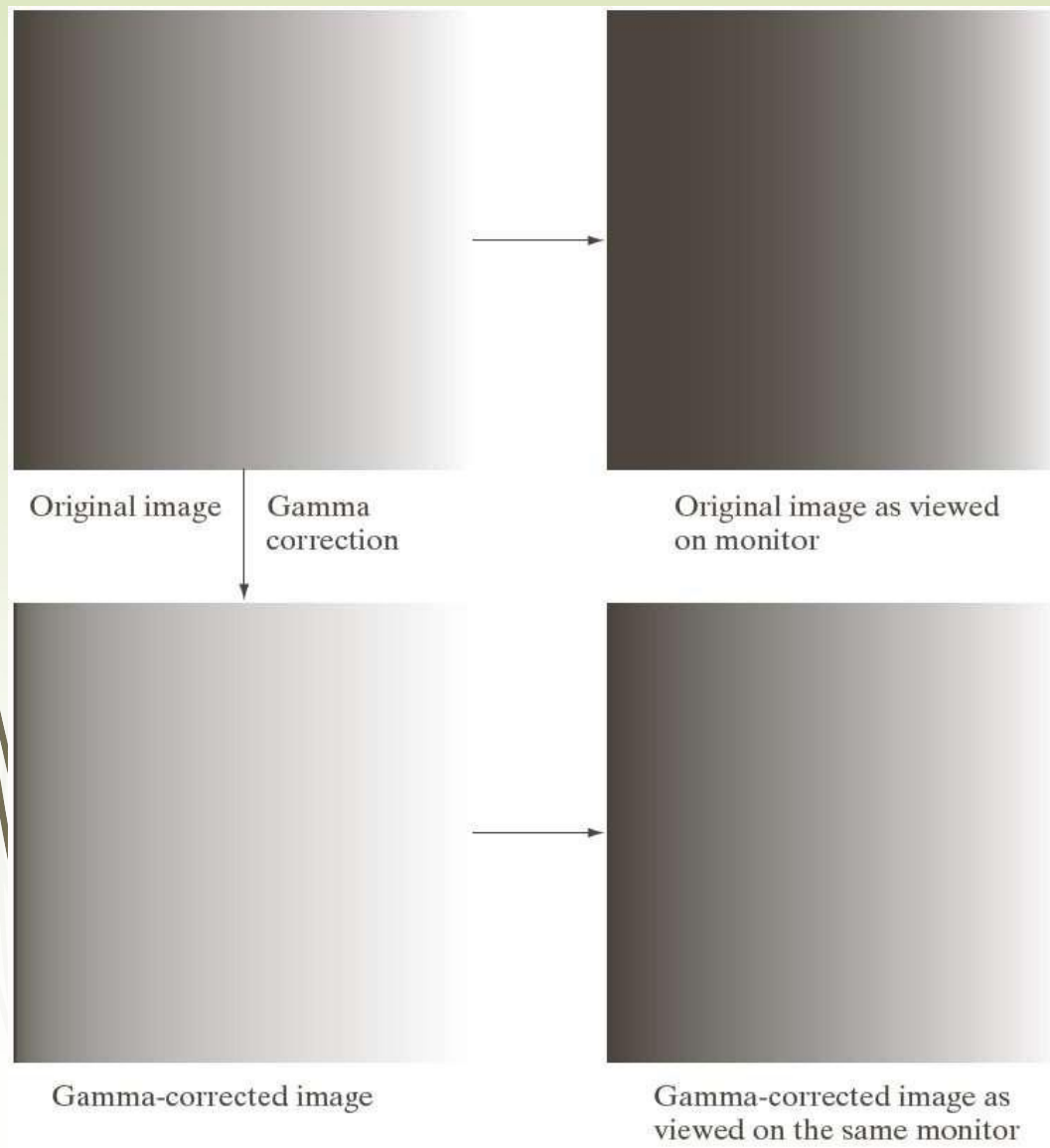


FIGURE 3.6 Plots of the equation $s = cr^\gamma$ for various values of γ ($c = 1$ in all cases). All curves were scaled to fit in the range shown.

Some Basic Intensity Transformation Functions



a	b
c	d

FIGURE 3.7

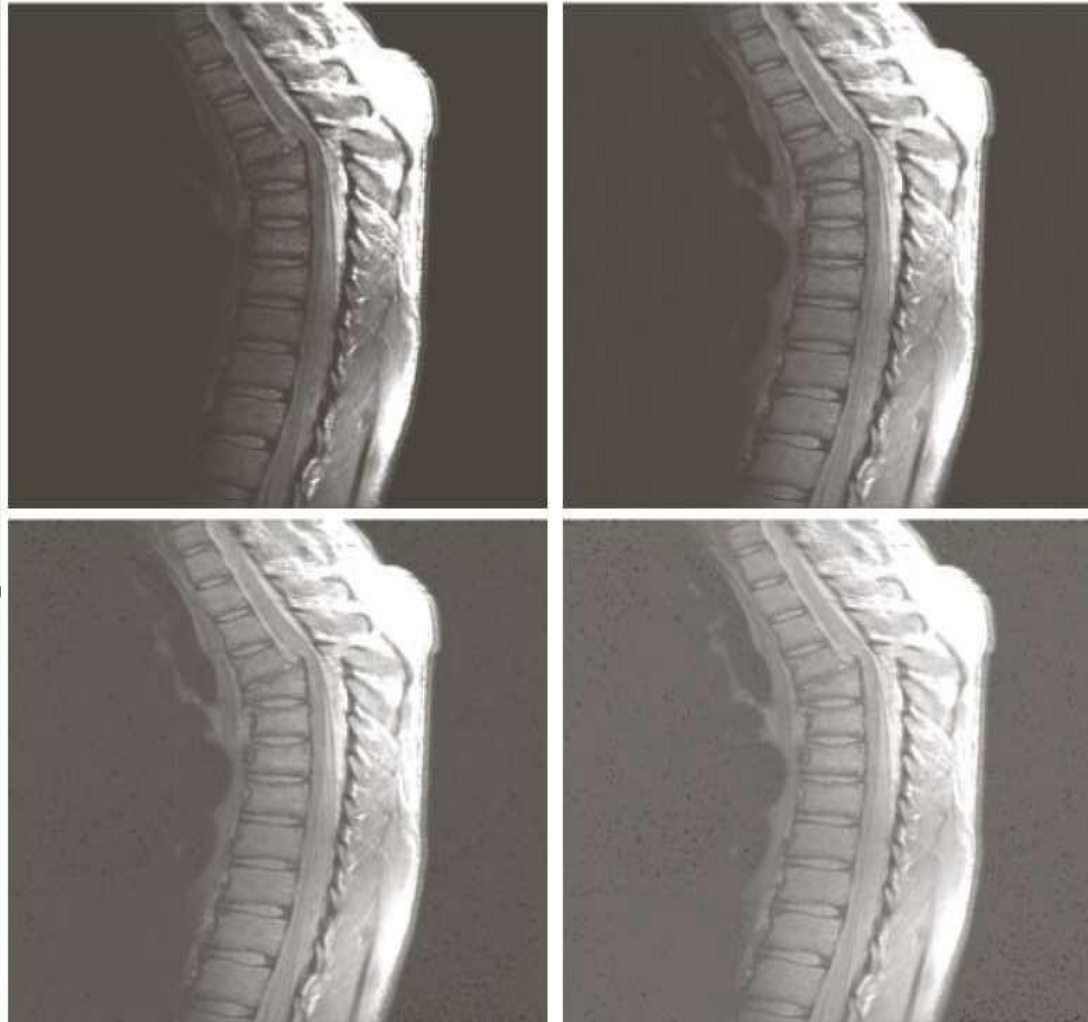
(a) Intensity ramp image. (b) Image as viewed on a simulated monitor with a gamma of 2.5. (c) Gamma-corrected image. (d) Corrected image as viewed on the same monitor. Compare (d) and (a).

Some Basic Intensity Transformation Functions

• Power Law (Gamma) Transformations

- Images that are not corrected properly look either bleached out or too dark.
- Varying gamma changes not only intensity, but also the ratio of red to green to blue in a color image.
- Gamma correction has become important, as the use of digital images over the internet is increasing.
- Useful for general purpose contrast manipulation.
- Apply gamma correction on CRT (Television, monitor), printers, scanners etc.
- Gamma value depends on device.

Some Basic Intensity Transformation Functions



a	b
c	d

FIGURE 3.8

(a) Magnetic resonance image (MRI) of a fractured human spine.

(b)–(d) Results of applying the transformation in Eq. (3.2-3) with $c = 1$ and $\gamma = 0.6, 0.4$, and 0.3 , respectively.

(Original image courtesy of Dr. David R. Pickens, Department of Radiology and Radiological Sciences, Vanderbilt University Medical Center.)

Some Basic Intensity Transformation Functions



a	b
c	d

FIGURE 3.9

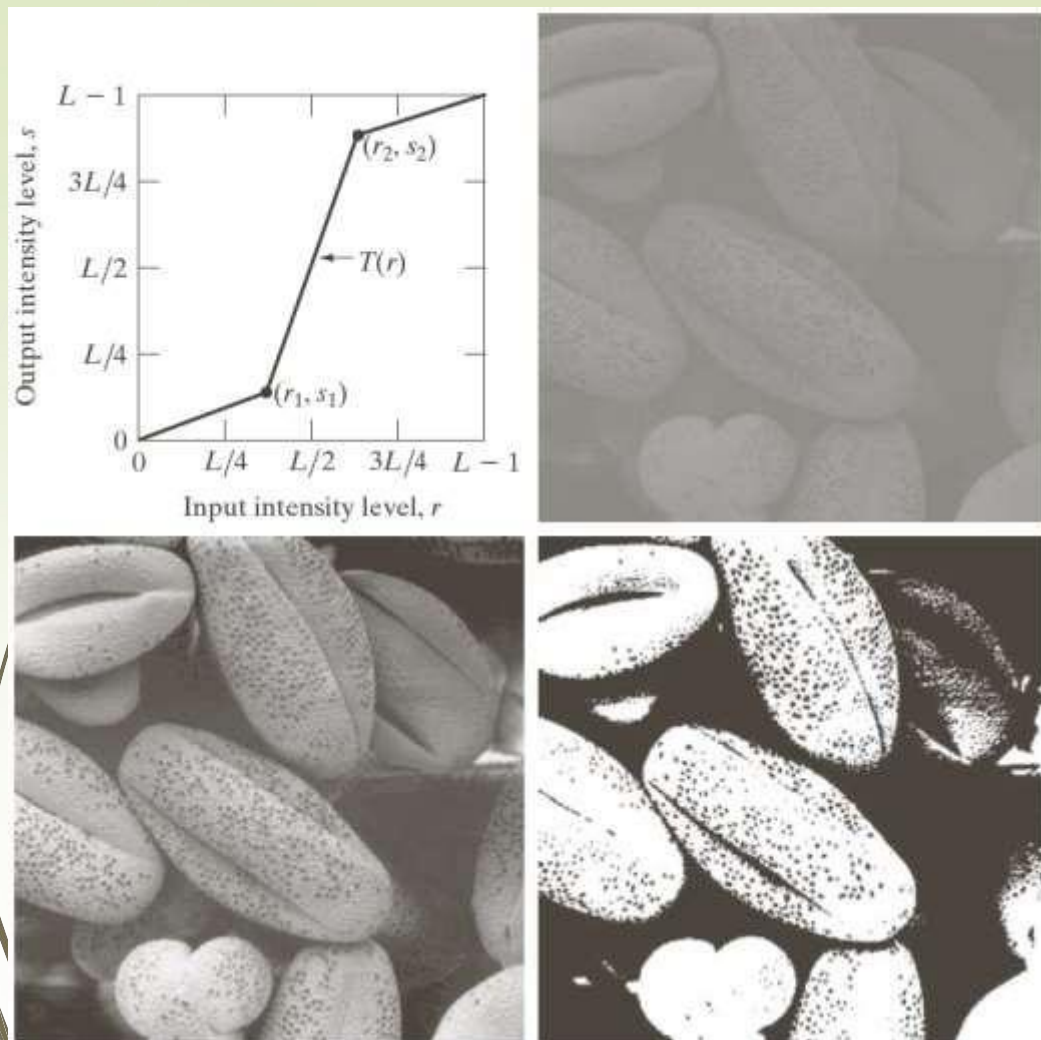
(a) Aerial image.
(b)–(d) Results of applying the transformation in Eq. (3.2-3) with $c = 1$ and $\gamma = 3.0, 4.0$, and 5.0 , respectively. (Original image for this example courtesy of NASA.)

Piecewise-Linear Transformation Functions

- Contrast Stretching

- Low contrast images can result from poor illuminations.
- Lack of dynamic range in the imaging sensor, or even the wrong setting of a lens aperture during image acquisition.
- It expands the range of intensity levels in an image so that it spans the full intensity range of display devices.
- Contrast stretching is obtained by setting $(r_1, s_1) = (r_{\min}, 0)$ and $(r_2, s_2) = (r_{\max}, L-1)$

Piecewise-Linear Transformation Functions



a	b
c	d

FIGURE 3.10

Contrast stretching.

(a) Form of

transformation

function. (b) A

low-contrast image.

(c) Result of

contrast stretching.

(d) Result of

thresholding.

(Original image

courtesy of Dr.

Roger Heady,

Research School of

Biological Sciences,

Australian National

University,

Canberra,

Australia.)

Piecewise-Linear Transformation Functions

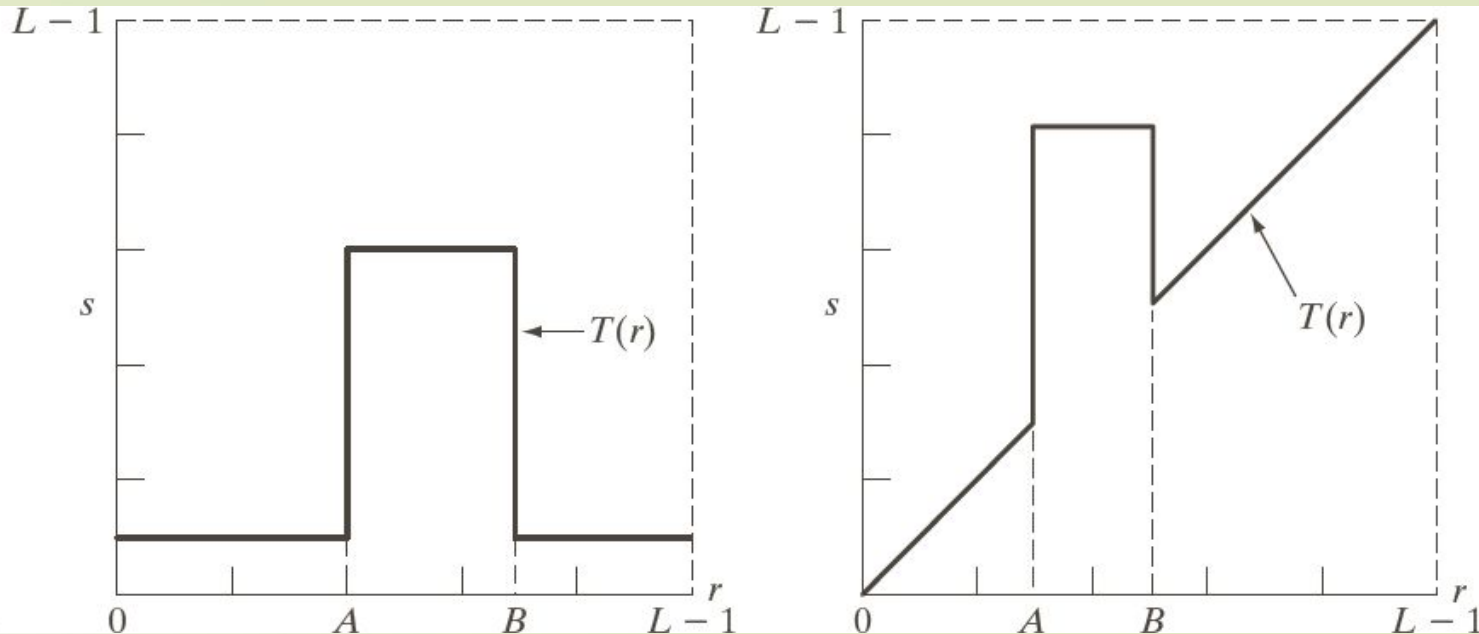
- Intensity Level Slicing
 - Highlighting specific range of intensities in an image.
 - Enhances features such as masses of water in satellite imagery and enhancing flaws in X-ray images.
 - It can be Implemented two ways:
 - 1) To display only one value (say, white) in the range of interest and rests are black which produces binary image.
 - 2) brightens (or darkens) the desired range of intensities but leaves all other intensity levels in the image unchanged.

Piecewise-Linear Transformation Functions

- Intensity Level Slicing

a b

FIGURE 3.11 (a) This transformation highlights intensity range $[A, B]$ and reduces all other intensities to a lower level. (b) This transformation highlights range $[A, B]$ and preserves all other intensity levels.



Piecewise-Linear Transformation Functions

- Intensity Level Slicing



a b c

FIGURE 3.12 (a) Aortic angiogram. (b) Result of using a slicing transformation of the type illustrated in Fig. 3.11(a), with the range of intensities of interest selected in the upper end of the gray scale. (c) Result of using the transformation in Fig. 3.11(b), with the selected area set to black, so that grays in the area of the blood vessels and kidneys were preserved. (Original image courtesy of Dr. Thomas R. Gest, University of Michigan Medical School.)

Piecewise-Linear Transformation Functions

- Bit Plane Slicing
 - Pixels are digital numbers composed of bits.
 - 256 gray scale image is composed of 8 bits.
 - Instead of highlighting intensity level ranges, we could highlight the contribution made to total image appearance by specific bits.
 - 8-bit image may be considered as being composed of eight 1-bit planes, with plane 1 containing the lowest-order bit of all pixels in the image and plane 8 all the highest-order bits.

Piecewise-Linear Transformation Functions

- Bit Plane Slicing

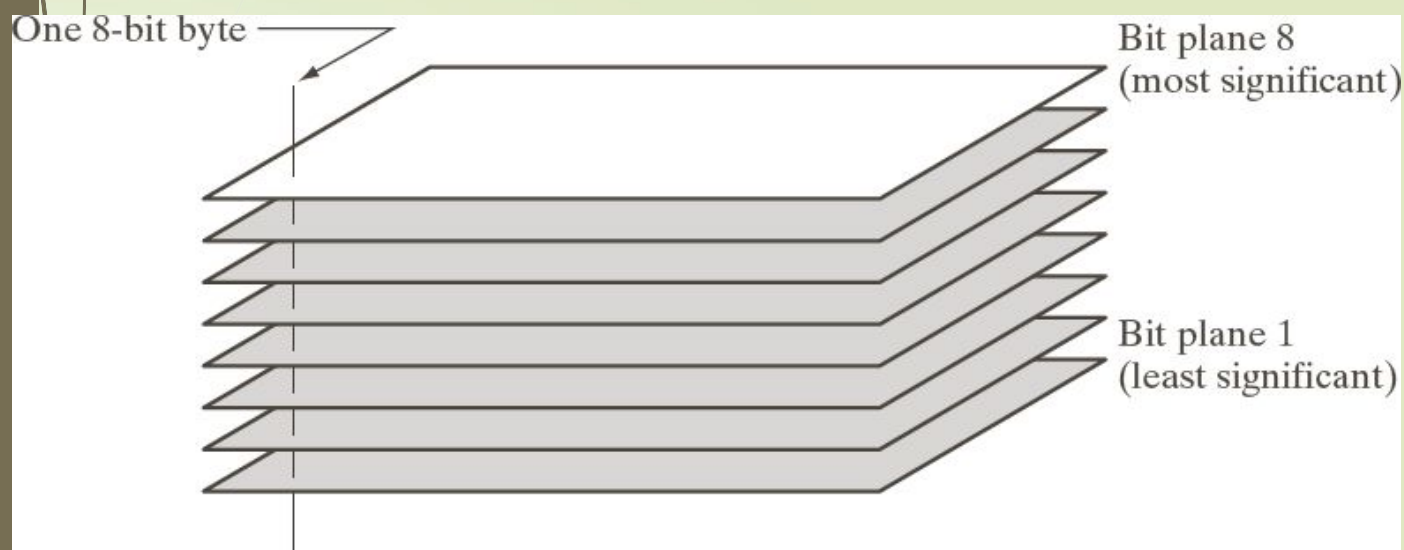


FIGURE 3.13
Bit-plane
representation of
an 8-bit image.

Piecewise-Linear Transformation Functions

- Bit Plane Slicing



FIGURE 3.14 (a) An 8-bit gray-scale image of size 500×1192 pixels. (b) through (i) Bit planes 1 through 8, with bit plane 1 corresponding to the least significant bit. Each bit plane is a binary image.

Piecewise-Linear Transformation Functions

- Bit Plane Slicing



a b c

FIGURE 3.15 Images reconstructed using (a) bit planes 8 and 7; (b) bit planes 8, 7, and 6; and (c) bit planes 8, 7, 6, and 5. Compare (c) with Fig. 3.14(a).

Histogram Processing

- Histogram of a digital image with intensity levels in the range $[0, L-1]$ is a discrete function $h(r_k) = n_k$, where r_k is the k th intensity value and n_k is the number of pixels in the image with intensity r_k
- Normalized histogram $p(r_k) = n_k / MN$, for $k = 0, 1, 2, \dots, L-1$.
- Histogram manipulation can be used for image enhancement.
- Information inherent in histogram also is quite useful in other image processing applications, such as image compression and segmentation.

Histogram Equalization

- Intensity mapping form

$$s = T(r), 0 \leq r \leq L - 1$$

Conditions:

a) $T(r)$ is a monotonically increasing function in the interval $[0, L-1]$ and

b) $0 \leq T(r) \leq L - 1$ for $0 \leq r \leq L - 1$

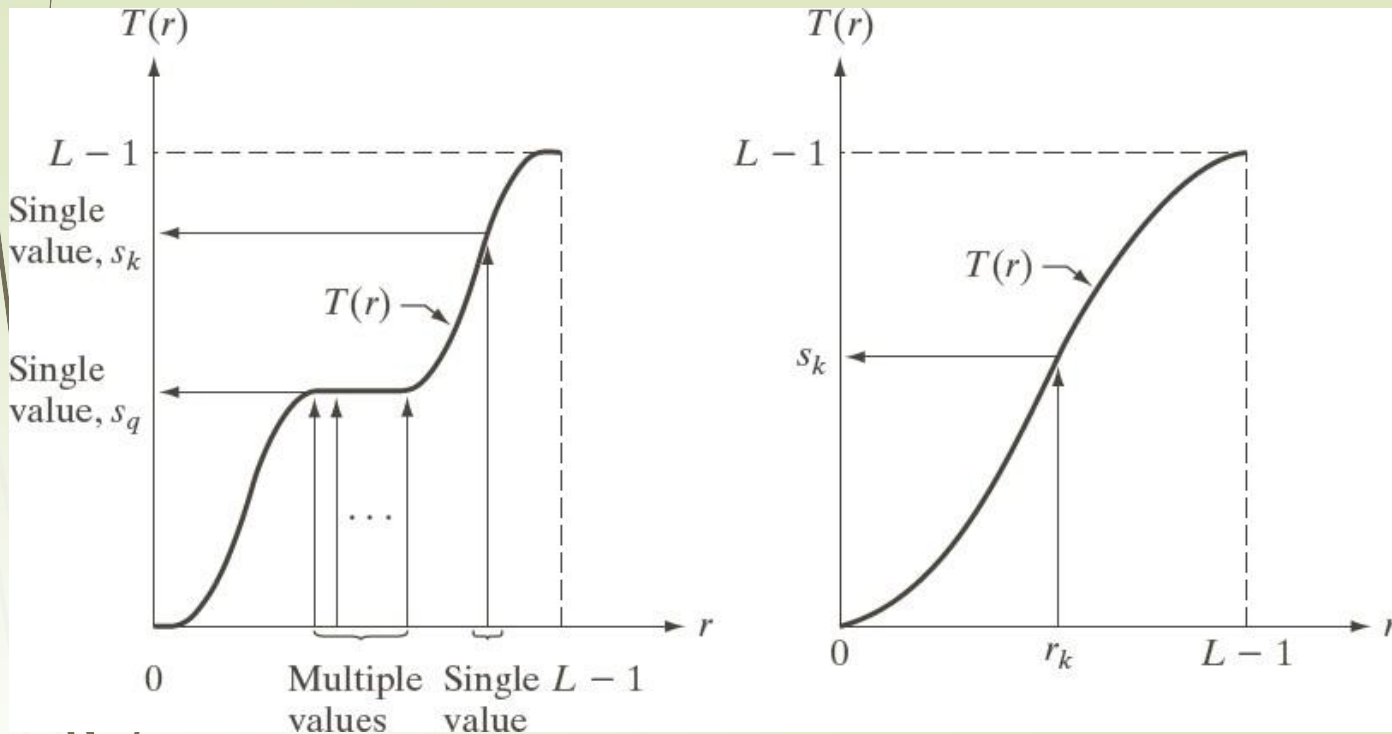
In some formulations, we use the inverse in which case

$$r = T^{-1}(s), 0 \leq s \leq 1$$

(a) change to

a') $T(r)$ is a strictly monotonically increasing function in the interval $[0, L-1]$

Histogram Processing

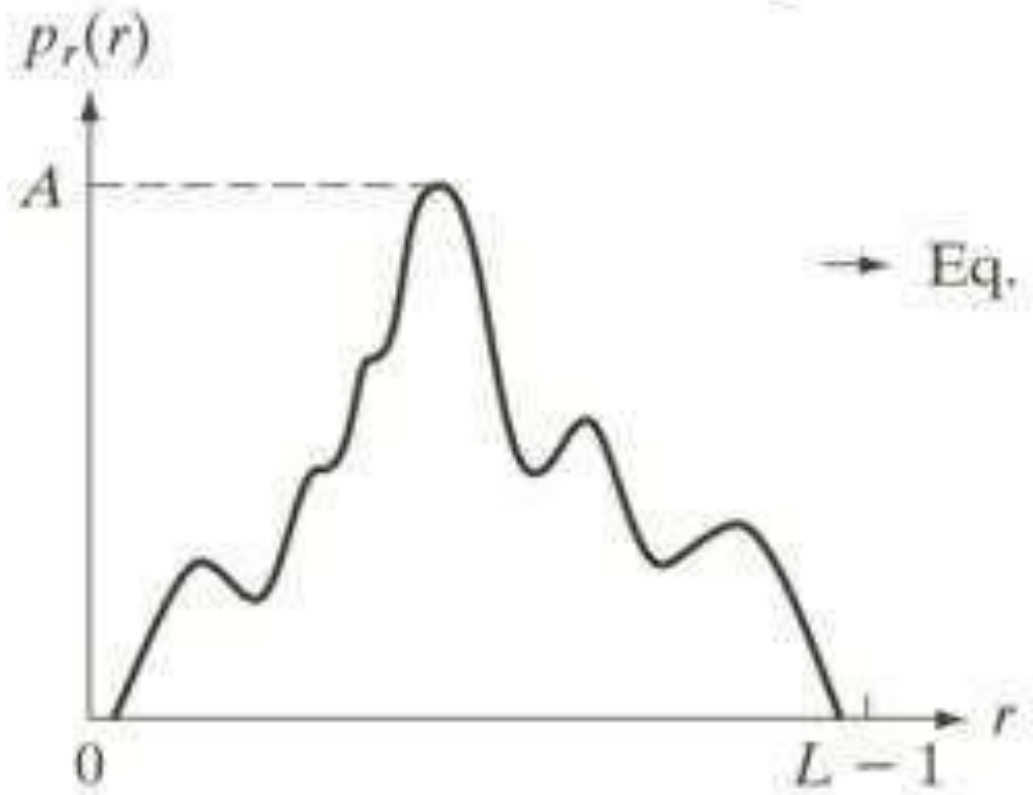
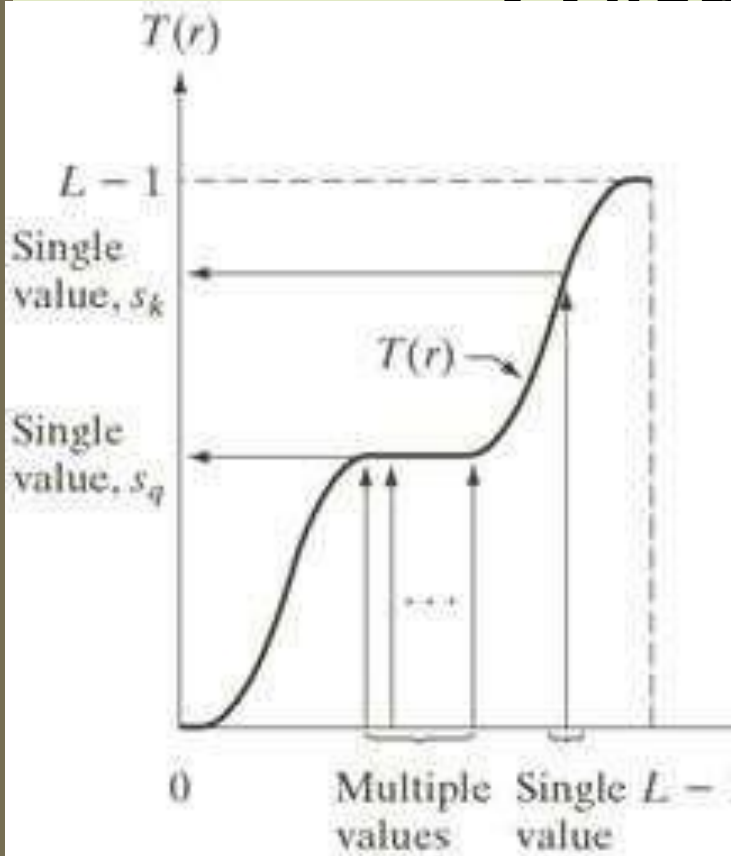


a b

FIGURE 3.17

(a) Monotonically increasing function, showing how multiple values can map to a single value. (b) Strictly monotonically increasing function. This is a one-to-one mapping, both ways.

Histogram Processing



Histogram Equalization

- Intensity levels in an image may be viewed as random variables in the interval $[0, L-1]$
- Fundamental descriptor of a random variable is its probability density function (PDF)
- Let $p_r(r)$ and $p_s(s)$ denote the PDFs of r and s respectively

$$p_s(s) = p_r(r) \left| \frac{dr}{ds} \right|$$

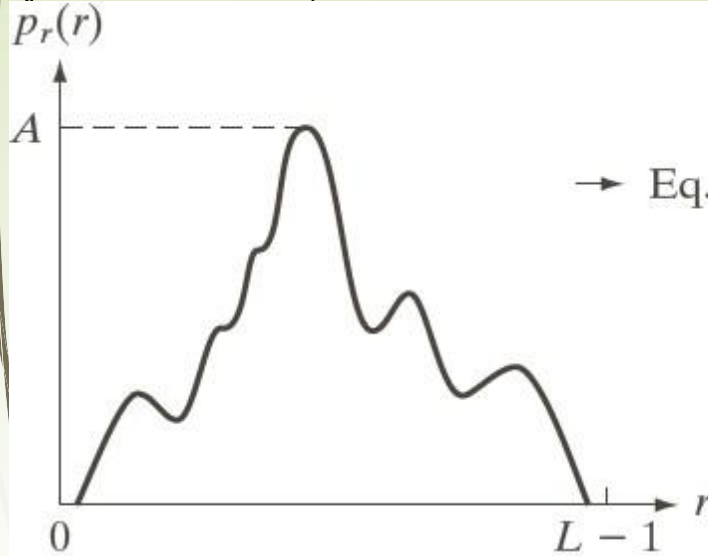
$$s = T(r) = (L-1) \int_0^r p_r(w) dw$$

$$\frac{ds}{dr} = \frac{dT(r)}{dr} = (L-1) \left[\frac{d}{dr} \int_0^r p_r(w) dw \right] = (L-1) p_r(r)$$

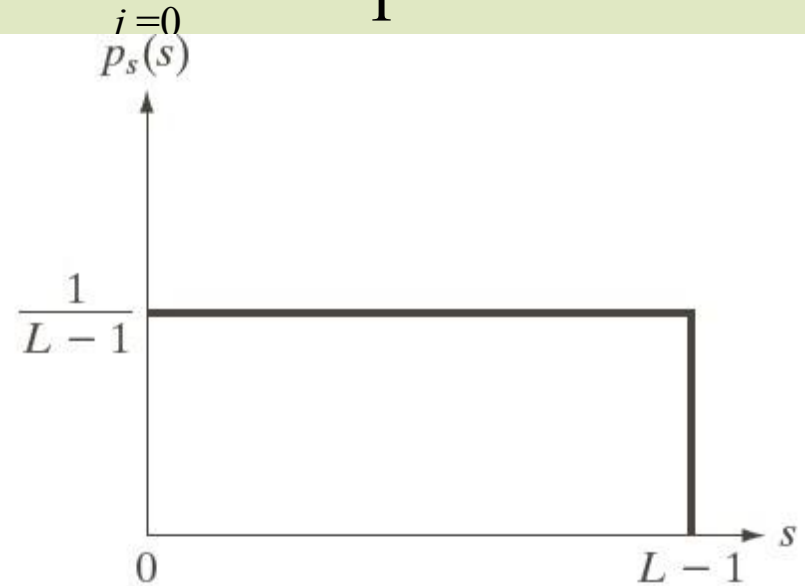
Histogram Equalization

$$p_s(s) = \frac{1}{L-1}$$

$$s_k = T(r_k) = (L-1) \sum_{j=0}^k \frac{p(r_j)}{MN-1}, \quad k = 0, 1, 2, \dots, L-1$$



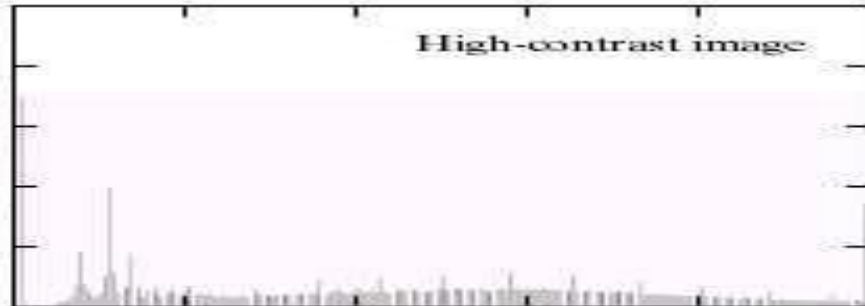
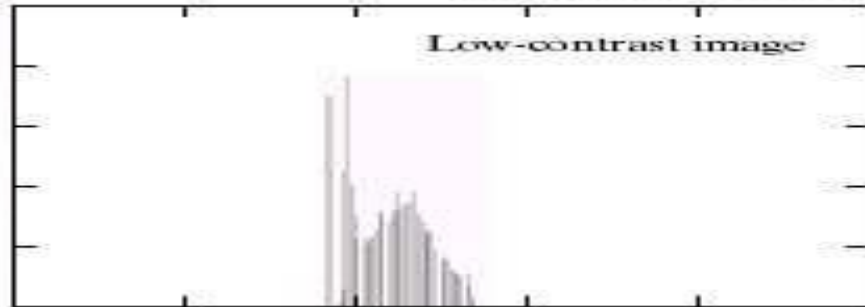
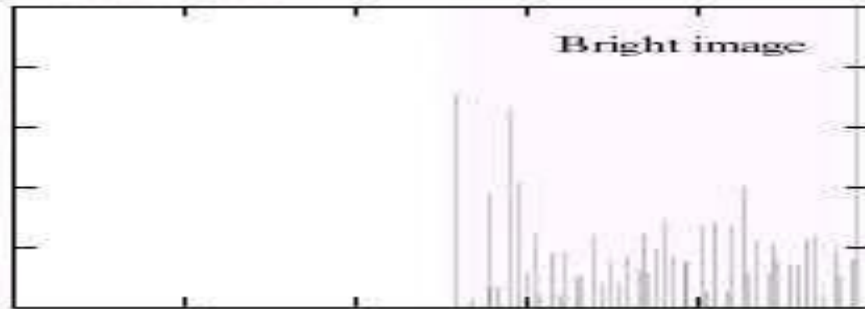
→ Eq. (3.3-4) →



a b

FIGURE 3.18 (a) An arbitrary PDF. (b) Result of applying the transformation in Eq. (3.3-4) to all intensity levels, r . The resulting intensities, s , have a uniform PDF, independently of the form of the PDF of the r 's.

Histogram



Histogram Equalization

r_k	n_k	$p_r(r_k) = n_k/MN$
$r_0 = 0$	790	0.19
$r_1 = 1$	1023	0.25
$r_2 = 2$	850	0.21
$r_3 = 3$	656	0.16
$r_4 = 4$	329	0.08
$r_5 = 5$	245	0.06
$r_6 = 6$	122	0.03
$r_7 = 7$	81	0.02

TABLE 3.1

Intensity distribution and histogram values for a 3-bit, 64×64 digital image.

Histogram Equalization

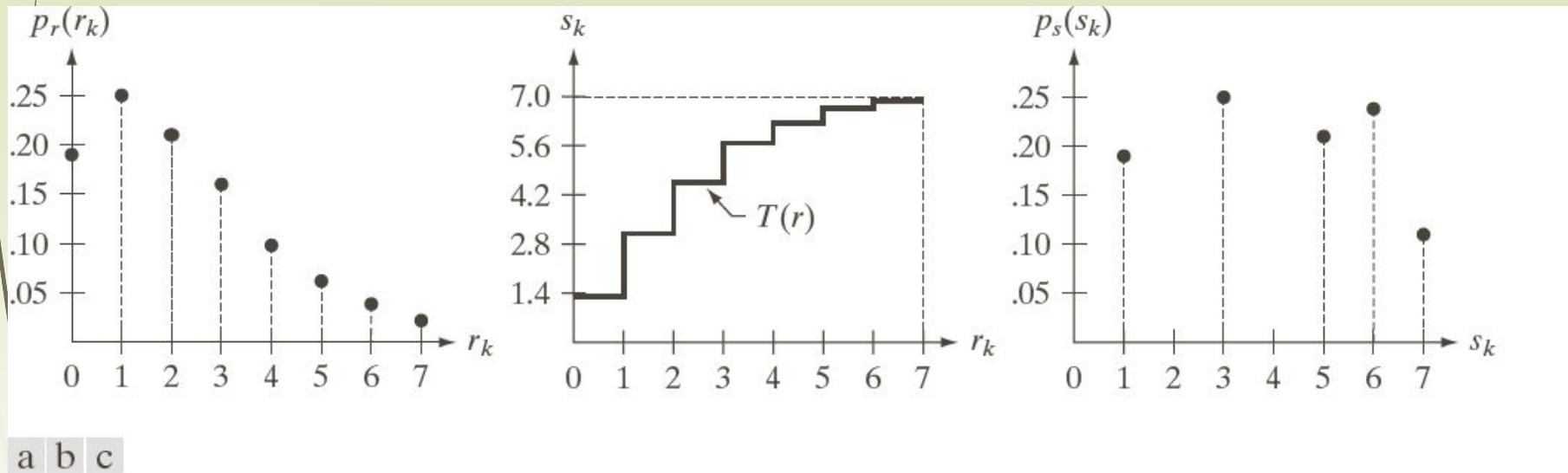


FIGURE 3.19 Illustration of histogram equalization of a 3-bit (8 intensity levels) image. (a) Original histogram. (b) Transformation function. (c) Equalized histogram.

Histogram Equalization

- Transformation functions

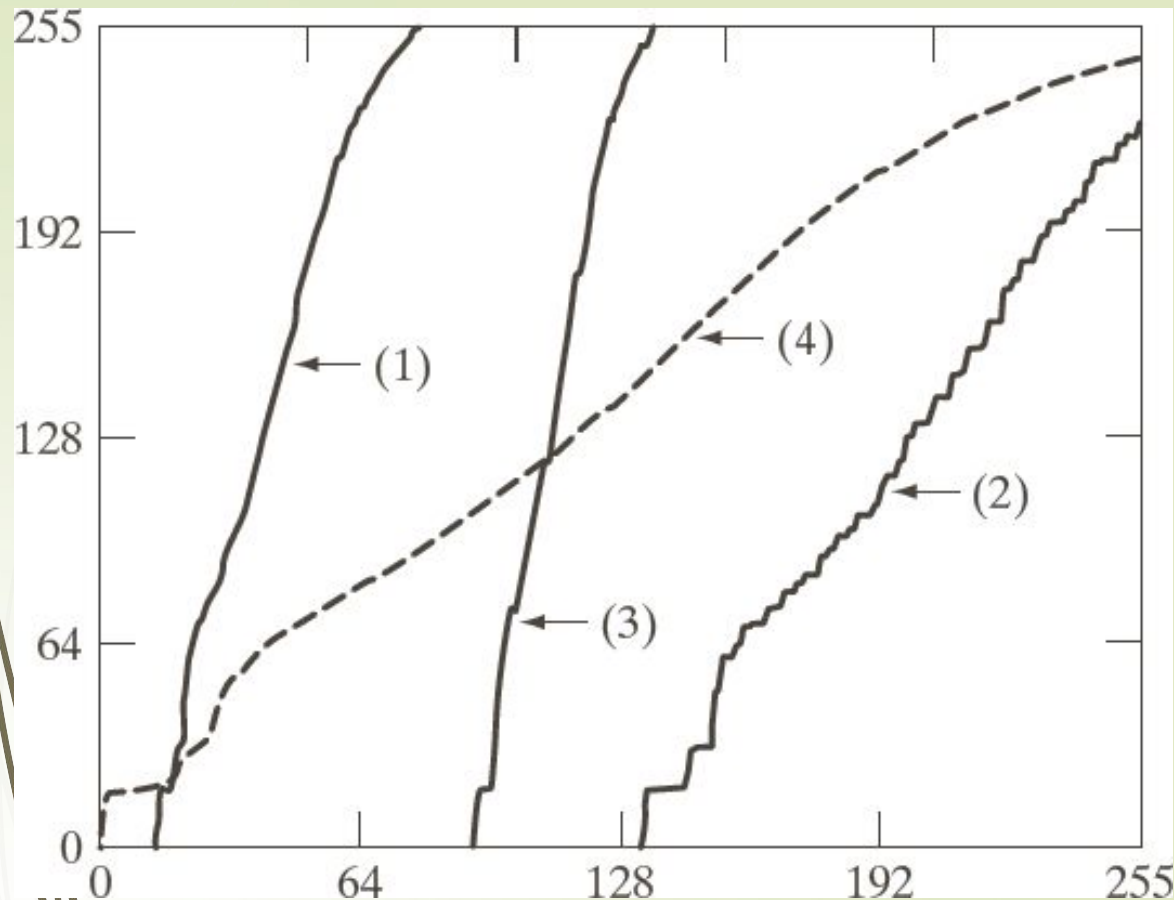


FIGURE 3.21

Transformation functions for histogram equalization. Transformations (1) through (4) were obtained from the histograms of the images (from top to bottom) in the left column of Fig. 3.20 using Eq. (3.3-8).

Histogram

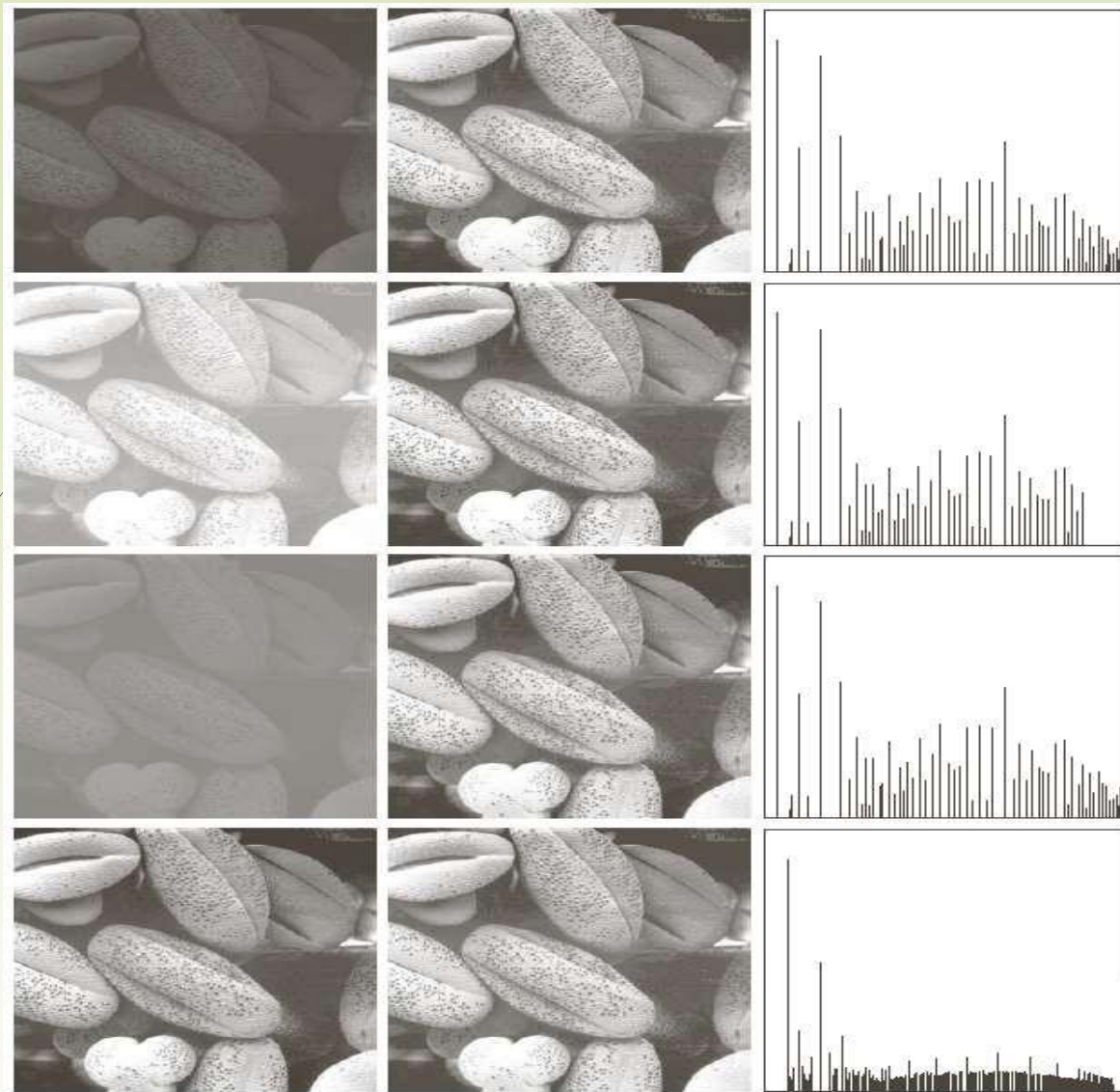


FIGURE 3.20 Left column: images from Fig. 3.16. Center column: corresponding histogram-equalized images. Right column: histograms of the images in the center column.

Histogram Matching (Specification)

- Histogram equalization automatically determines a transformation function produce uniform histogram
- When automatic enhancement is desired, equalization is a good approach
- There are some applications in which attempting to base enhancement on a uniform histogram is not the best approach
- In particular, it is useful sometimes to be able to specify the shape of the histogram that we wish the processed image to have.
- The method used to generate a processed image that has a specified histogram is called *histogram matching* or *specification*

Histogram Matching (Specification)

- Histogram Specification Procedure:

1) Compute the histogram $p_r(r)$ of the given image, and use it to find the histogram equalization transformation equation

$$s_k = T(r_k) = (L-1) \sum_{j=0}^k \frac{n_j}{MN}, \quad k = 0, 1, 2, \dots, L-1$$

and round the resulting values to the integer range $[0, L-1]$

2) Compute all values of the transformation function G using same equation

and round values of G

$$G(z_q) = (L-1) \sum_{i=0}^q p_z(r_i), \quad q = 0, 1, 2, \dots, L-1$$

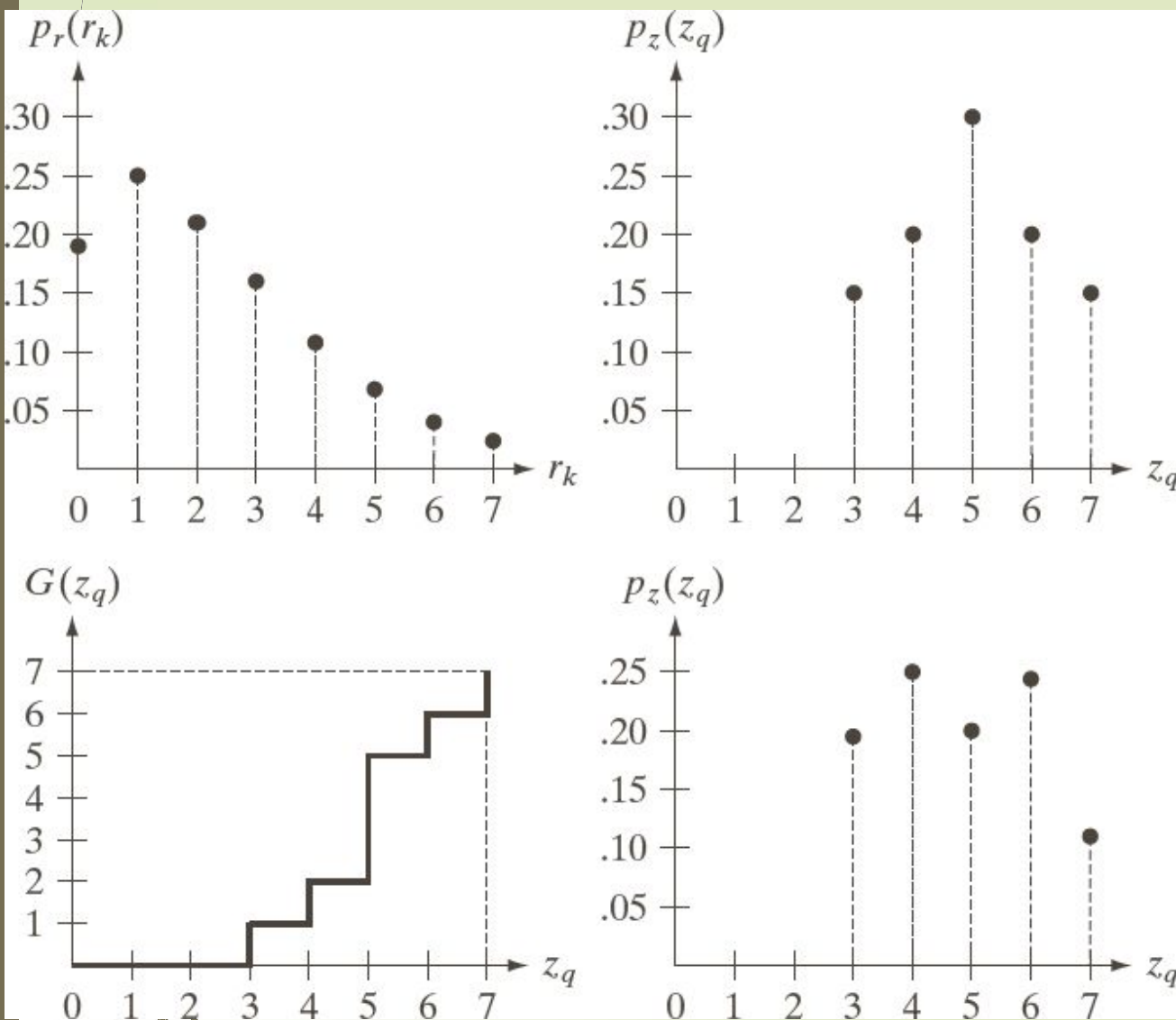
3) For every value of s_k , $k = 0, 1, \dots, L-1$, use the stored values of G to find the corresponding value of z_q so that $G(z_q)$ is closet to s_k and store these mappings from s to z .

Histogram Matching (Specification)

- Histogram Specification Procedure:

4) Form the histogram-specified image by first histogram-equalizing the input image and then mapping every equalized pixel value, s_k , of this image to the corresponding value z_q in the histogram-specified image using the mappings found in step 3.

Histogram Matching



a	b
c	d

FIGURE 3.22

(a) Histogram of a 3-bit image. (b)

Specified histogram.

(c) Transformation function obtained from the specified histogram.

(d) Result of performing histogram specification. Compare (b) and (d).

Histogram Matching

z_q	Specified $p_z(z_q)$	Actual $p_z(z_k)$
$z_0 = 0$	0.00	0.00
$z_1 = 1$	0.00	0.00
$z_2 = 2$	0.00	0.00
$z_3 = 3$	0.15	0.19
$z_4 = 4$	0.20	0.25
$z_5 = 5$	0.30	0.21
$z_6 = 6$	0.20	0.24
$z_7 = 7$	0.15	0.11

TABLE 3.2

Specified and actual histograms (the values in the third column are from the computations performed in the body of Example 3.8).

Histogram Matching

z_q	$G(z_q)$
$z_0 = 0$	0
$z_1 = 1$	0
$z_2 = 2$	0
$z_3 = 3$	1
$z_4 = 4$	2
$z_5 = 5$	5
$z_6 = 6$	6
$z_7 = 7$	7

TABLE 3.3

All possible values of the transformation function G scaled, rounded, and ordered with respect to z .

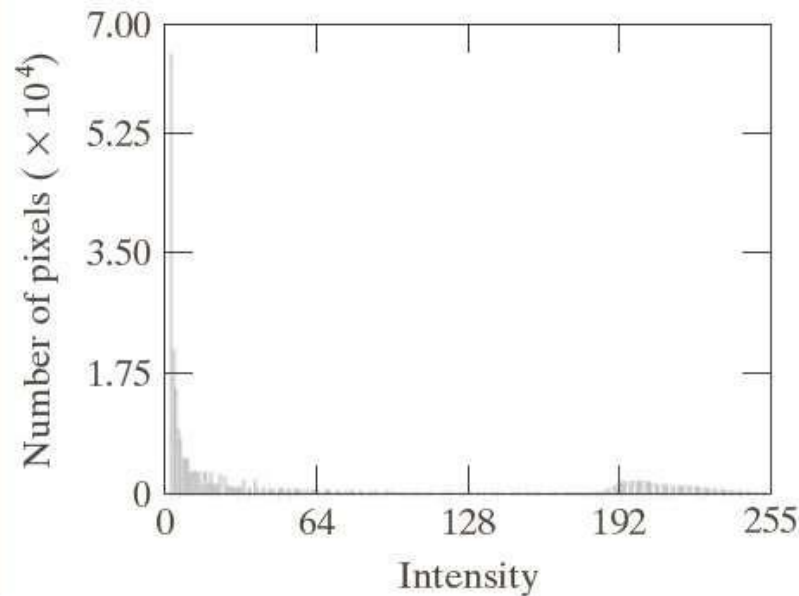
r_k	s_k	$G(z_q)$
0	1	0
1	3	0
2	5	0
3	6	1
4	6	2
5	7	5
6	7	6
7	7	7

s_k	$\rightarrow$	z_q
1	$\rightarrow$	3
3	$\rightarrow$	4
5	$\rightarrow$	5
6	$\rightarrow$	6
7	$\rightarrow$	7

TABLE 3.4

Mappings of all the values of s_k into corresponding values of z_q .

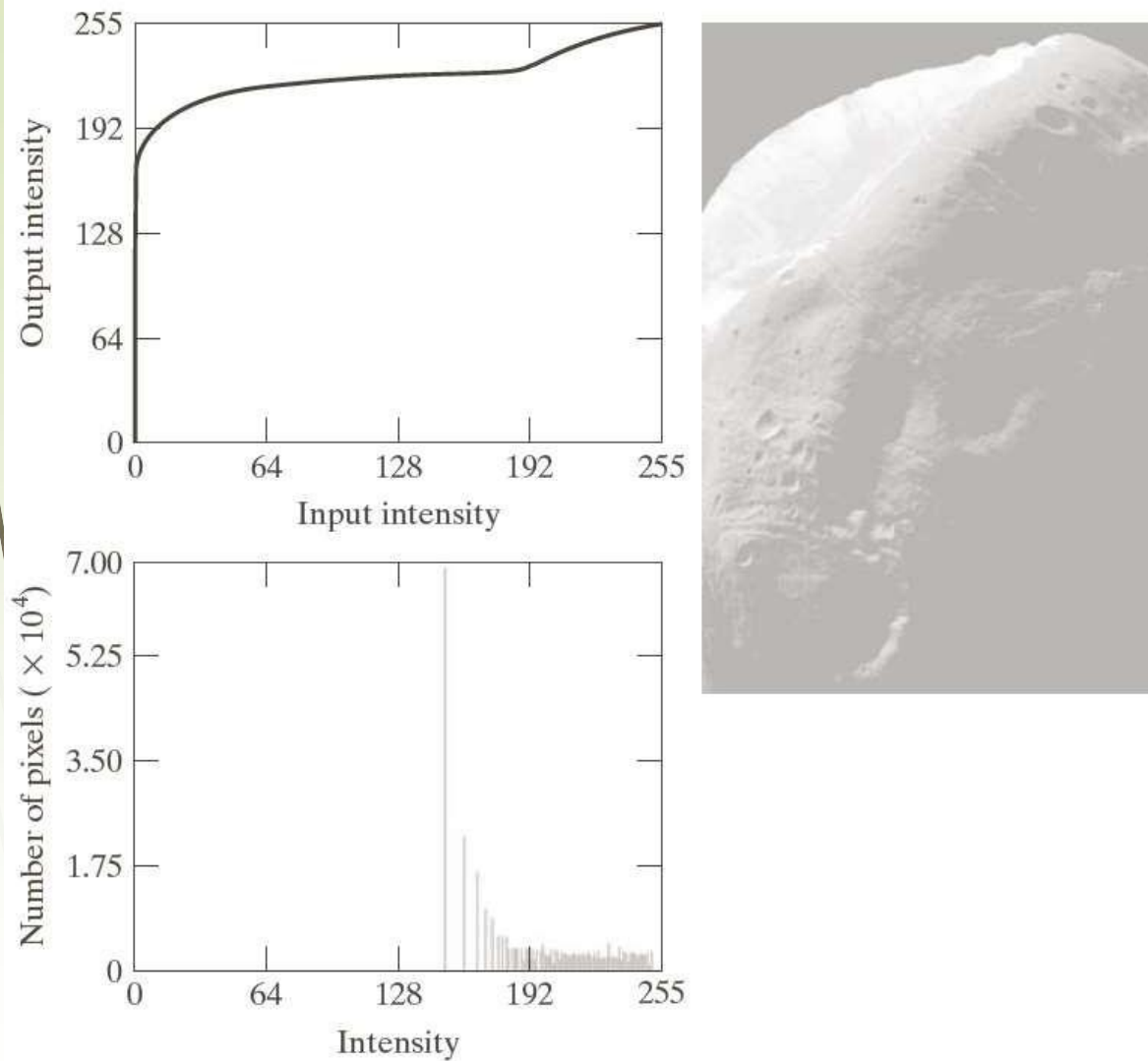
Histogram Matching



a b

FIGURE 3.23
(a) Image of the Mars moon Phobos taken by NASA's *Mars Global Surveyor*.
(b) Histogram.
(Original image courtesy of NASA.)

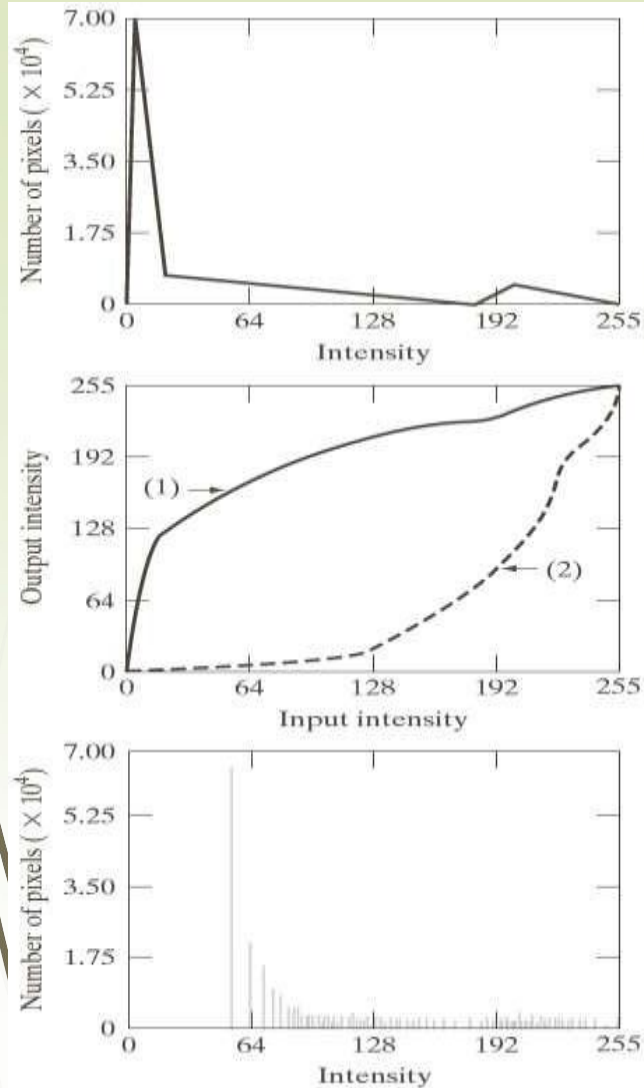
Histogram Matching



a b
c

FIGURE 3.24
(a) Transformation function for histogram equalization.
(b) Histogram-equalized image (note the washed-out appearance).
(c) Histogram of (b).

Histogram Matching



a c
b
d

FIGURE 3.25

(a) Specified histogram.

(b) Transformations.
(c) Enhanced image using mappings from curve (2).

(d) Histogram of (c).

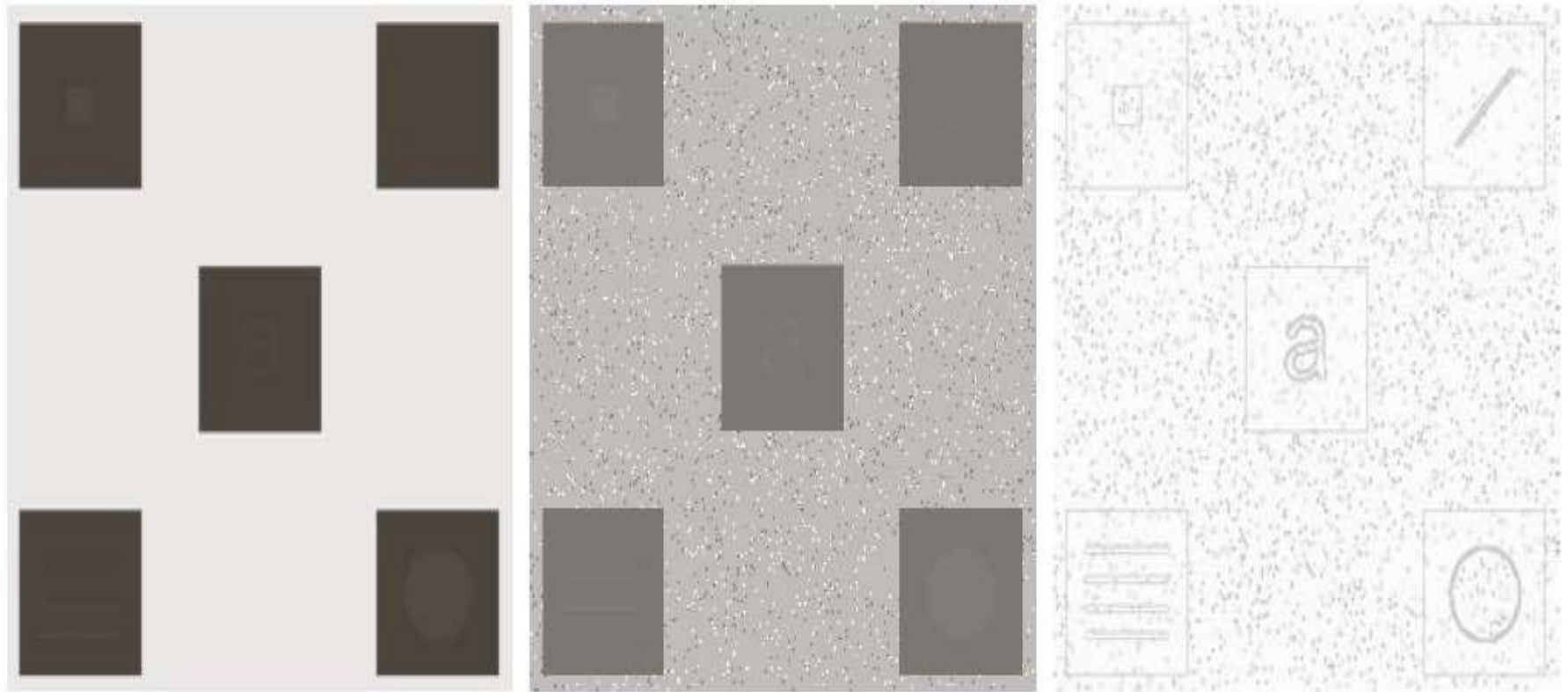
Local Histogram Processing

- Histogram Processing methods discussed in the previous two sections are *Global*, in the sense that pixels are modified by a transformation function based on the intensity distribution of an entire image.
- There are some cases in which it is necessary to enhance detail over small areas in an image.
- This procedure is to define a neighborhood and move its center pixel to pixel.
- At each location, the histogram of the points in the neighborhood is computed and either a histogram equalization or histogram specification transformation function is obtained.
- Map the intensity of the pixel centered in the neighborhood.
- Center of the neighborhood region is then moved to an adjacent pixel location and the procedure is repeated.

Local Histogram Processing

- This approach has obvious advantages over repeatedly computing the histogram of all pixels in the neighborhood region each time the region is moved one pixel location.
- Another approach used sometimes to reduce computation is to utilize non overlapping regions, but this method usually produces an undesirable “*blocky*” effect.

Local Histogram Processing



a b c

FIGURE 3.26 (a) Original image. (b) Result of global histogram equalization. (c) Result of local histogram equalization applied to (a), using a neighborhood of size 3×3 .

Acknowledgment

- “Digital Image Processing”, 2/ E, Rafael C. Gonzalez & Richard E. Woods,

